STAT 847 Final Project

AI Premium Effect Validation in U.S. IT Labor Markets

Research Focus and Dataset Sources

My project focuses on one central idea: **does higher AI demand penetration produce a measurable wage premium in regional IT labor markets?**

In this report, AI demand penetration is measured using state-level AI job posting intensity, while the wage baseline is measured using annual mean and median wages for Computer and Mathematical Occupations.

The combined panel in this project is constructed from two public source systems: *BLS OEWS* and the *Stanford AI Index* public data releases. The following table lists the source endpoints.

Dataset	Component	Year(s)	Source URL / Note
BLS OEWS	Computer and Mathematical Occupations (state estimates, all industries)	2020–2024	https://www.bls.gov/oes/special-requests/oesm20st.zip, https://www.bls.gov/oes/special-requests/oesm21st.zip, https://www.bls.gov/oes/special-requests/oesm22st.zip, https://www.bls.gov/oes/special-requests/oesm23st.zip, https://www.bls.gov/oes/special-requests/oesm24st.zip. Filter rule: `AREA_TYPE=2, NAICS=000000, OCC_CODE=15-0000`.
Stanford AI Index	AI Job Postings by State (workbook)	2020–2021	https://docs.google.com/spreadsheets/d/19UGjvQW-PcnWoHEPe9Wd1Eib5z6S5II7lu1tE92fK3M/export?format=xlsx. 2022 AI Index public data, sheet "AI Job Postings by State".
Stanford AI Index	AI job postings by state (CSV)	2022	https://drive.google.com/uc?export=download&id=1WRcpKKYZn6GHJQ2FBH1Xv-w_hgcOYrhV.
Stanford AI Index	Share of state job postings in AI (CSV)	2022	https://drive.google.com/uc?export=download&id=1MTYGYe4tdTXXRjWzQKcWWNDGngRtZXLt.
Stanford AI Index	AI job postings by state (CSV)	2023	https://drive.google.com/uc?export=download&id=12u4KhwxMPTN511UXMuLd0JBYsobkCWbV.
Stanford AI Index	Share of state job postings in AI (CSV)	2023	https://drive.google.com/uc?export=download&id=1pQdnT4vPfjs69yQfktiEiwXCc3rON8jF.
Stanford AI Index	AI job postings by state (CSV)	2024	https://drive.google.com/uc?export=download&id=1j-3F7QYncgYxXFp25Axgc2P3hRQb-KSj.
Stanford AI Index	Share of state job postings in AI (CSV)	2024	https://drive.google.com/uc?export=download&id=1QVXtUs0JBxHH3BYIz3S-3G9Zdwp7DIKp.

Exploration E3: ggpairs of Six Important Variables

I use the following six variables:

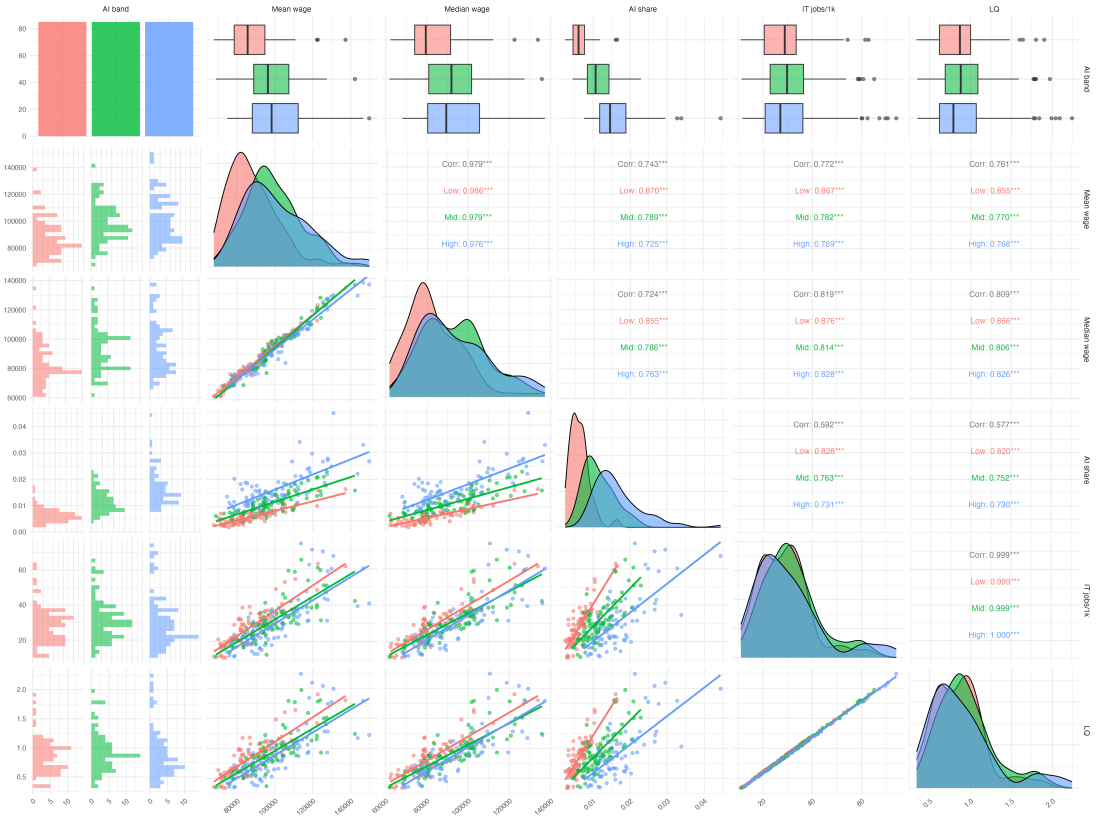
- Categorical: AI penetration band (`Low/Mid/High`), defined by terciles of AI postings per 1,000 IT workers.
- Continuous: mean annual wage, median annual wage, AI share of state postings, IT jobs per 1,000 workers, IT location quotient.

These six variables were chosen because they map directly to the core question in this project: whether stronger AI demand is accompanied by a higher regional IT wage baseline. Instead of looking at one variable at a time, the ggpairs matrix makes it possible to inspect many pairwise relationships at once and see whether the patterns are consistent.

The main visual findings are:

- Mean wage and median wage are strongly positively related.
- AI share and wage variables are positively related in raw pairwise views, but with visible dispersion.
- IT jobs per 1,000 and location quotient also trend positively with wages.
- High AI band points concentrate more in higher-wage regions, but overlap with Mid band is non-trivial.

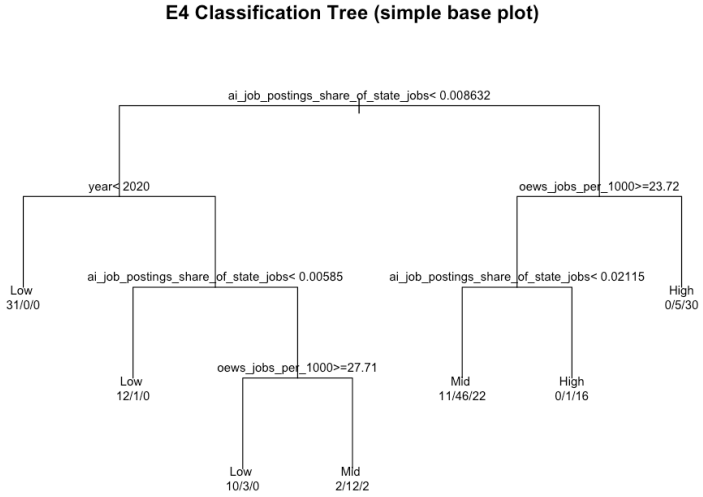
Taken together, the plot supports a cautious "yes" at the exploratory stage: high-AI observations appear more often in higher-wage parts of the panel. At the same time, the overlap between Mid and High categories is still substantial, which is why I treat E3 as pattern discovery rather than proof. In other words, the chart gives motivation for deeper modeling, but not a causal conclusion by itself.



Exploration E4: Classification Tree (Simple/Ugly Plot)

For E4, the response variable is `AI band`. Predictors are the other five variables from E3, and I also include `year` to capture basic time movement. The tree is intentionally simple and uses a base-R style plot, matching the assignment request for an "ugly" but interpretable classifier.

```
tree_model <- rpart(
  ai_intensity_band ~ oews_mean_annual_wage +
    oews_median_annual_wage +
    ai_job_postings_share_of_state_jobs +
    oews_jobs_per_1000 +
    oews_loc_quotient + year,
  data = train_set,
  method = "class",
  control = rpart.control(cp = 0.01, minsplit = 12, maxdepth = 4)
)
```



The confusion matrix below summarizes out-of-sample performance. Overall accuracy is 0.647, which is acceptable for a three-class split with overlapping classes.

Actual / Predicted	Low	Mid	High
Low	17	2	0
Mid	3	12	2
High	0	11	4

To satisfy the "follow two predictions down the tree" requirement, I traced two test cases:

- Example 1 (AR, 2020): Actual = Low, Predicted = Low. Path: $ai_job_postings_share_of_state_jobs < 0.009$ (left) $\rightarrow$ $year < 2020.5$ (left).
- Example 2 (CA, 2020): Actual = Mid, Predicted = Mid. Path: $ai_job_postings_share_of_state_jobs \geq 0.009$ (right) $\rightarrow$ $oews_jobs_per_1000 \geq 23.718$ (right).

Analysis A1: Necessary Value Transformations

For A1, I checked whether value transformation is necessary before running stronger models. Wage and AI penetration variables are both right-skewed, so I use:

$$\log(\text{mean wage}), \quad \log(\text{median wage}), \quad \log(1 + \text{AI penetration})$$

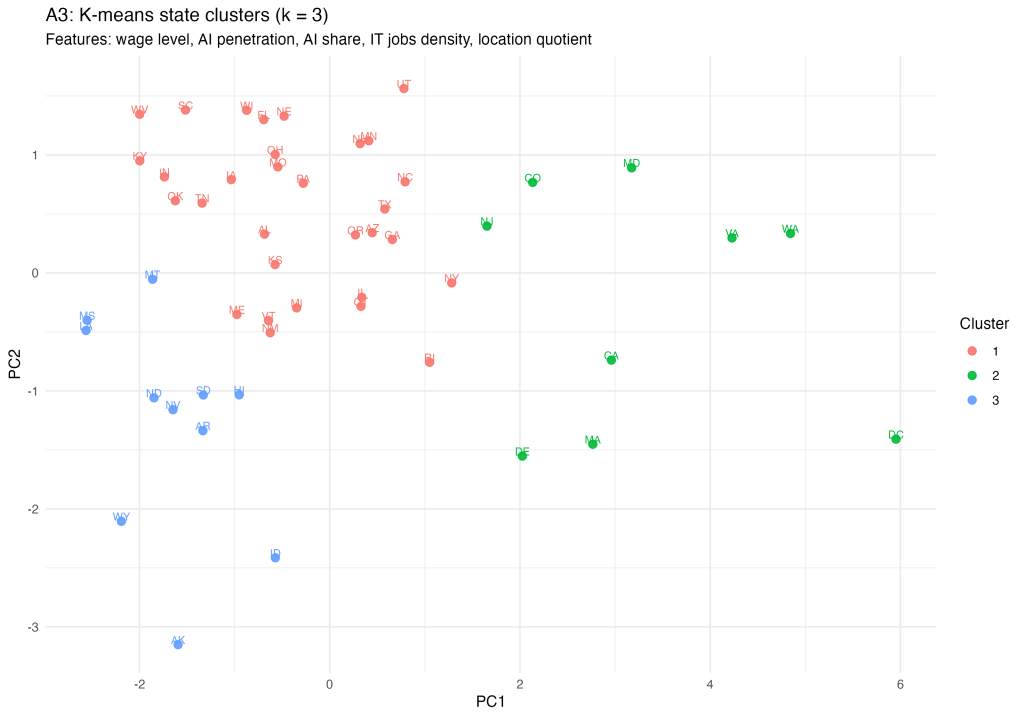
Posted diagnostics:

Metric	Value
Skewness(mean annual wage)	0.761
Skewness(log mean annual wage)	0.362
Skewness(residuals raw model)	0.917
Skewness(residuals log model)	0.347
Adj R^2 raw model	0.712
Adj R^2 log model	0.726

The changes are directionally clear: both variable skewness and residual skewness drop, and adjusted R^2 increases slightly. This does not radically change the story, but it makes later regressions better behaved and easier to interpret. For that reason, transformed specifications are carried into subsequent analysis sections.

Analysis A3: Clustering State Labor-Market Profiles

For A3, I run K-means clustering ($k = 3$, standardized features) on state-level averages of mean wage, AI penetration, AI share, IT jobs per 1,000, and location quotient. I use this step to test whether states naturally separate into distinct labor-market profiles, rather than forcing one pooled interpretation for all regions.



Cluster sizes:

Cluster	Number of states
1	31
2	9
3	11

The grouping pattern is meaningful: one large middle cluster and two smaller edge clusters. This supports the idea that the AI premium question is heterogeneous by state context. A single national slope can hide important differences between mature tech markets and thinner IT labor markets.

Analysis A5: Sophisticated Continuous-Response Model

In A5, I fit a two-way fixed-effects model with log(mean annual wage) as the response:

$$\log(wage_{s,t}) = \beta_1 \log(1 + AIpen_{s,t}) + \beta_2 JobsPer1k_{s,t} + \beta_3 LQ_{s,t} + \alpha_s + \gamma_t + \varepsilon_{s,t}.$$

I prefer this over pooled OLS because state fixed effects absorb persistent cross-state differences (industrial mix, local wage level, institutional context), while year fixed effects absorb macro shocks shared by all states.

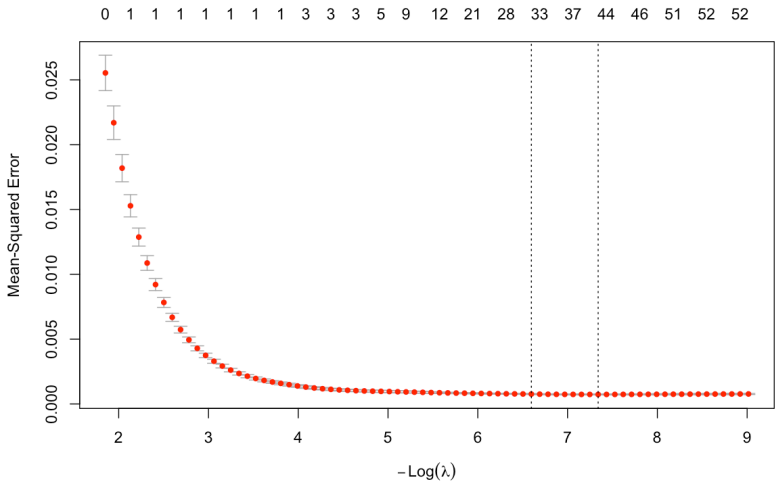
Key posted coefficients:

Term	Estimate	Std. Error	t value	p value
log_ai_penetration	-0.0115	0.0097	-1.191	0.235
oews_jobs_per_1000	-0.0187	0.0117	-1.604	0.110
oews_loc_quotient	0.7583	0.3806	1.992	0.048

With these controls, adjusted R^2 is high (≈ 0.967). The AI penetration coefficient is small and statistically insignificant, while location quotient remains significant at the 5% level. The practical reading is that broad state-level AI intensity alone does not explain wage changes once structural state and time effects are absorbed.

Analysis A6: Feature Extraction with LASSO

In A6, I apply LASSO with cross-validation for log(mean wage) as a feature extraction exercise. This complements A5 by focusing on predictive selection rather than coefficient testing.



Top selected features:

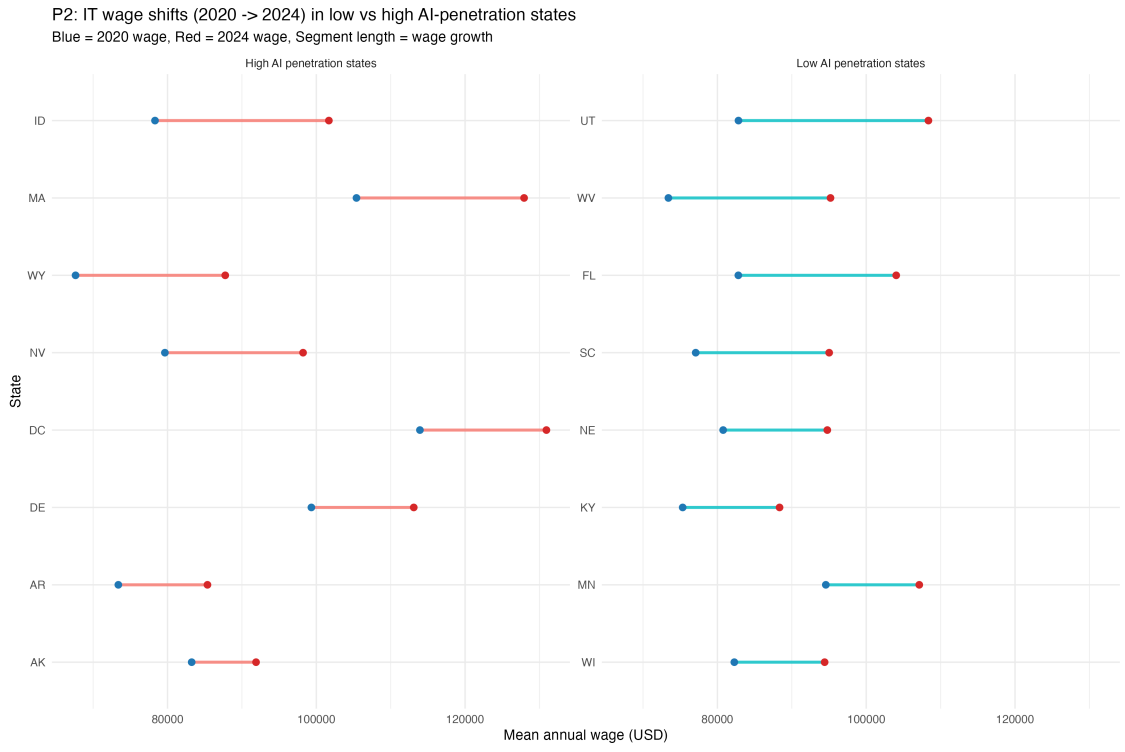
Feature	Coefficient
log_median_wage	0.7746
state_abbrevCA	0.0623
state_abbrevNY	0.0556
state_abbrevLA	0.0394
state_abbrevID	0.0391
state_abbrevAR	-0.0377
state_abbrevNE	-0.0343
state_abbrevND	-0.0308
ai_job_postings_share_of_state_jobs	0.0300
state_abbrevDC	-0.0268

The selected set shows that wage structure and persistent state effects dominate prediction, while AI share is still retained as a useful signal. This is consistent with A5: AI-related variables matter, but they are part of a broader system rather than a standalone mechanism.

An additional advantage of this step is robustness: LASSO shrinks weaker terms and reduces overfitting risk when many state indicators are included. So even though this is not a causal method, it is a useful check that the earlier regression story is not driven by one unstable specification.

Presentation P2: Novel Visualization

For P2, I use a dumbbell plot comparing 2020 and 2024 wages in the top-8 and bottom-8 states by average AI penetration. This design is tailored to this dataset because it highlights both level differences and change-over-time in one compact visual.

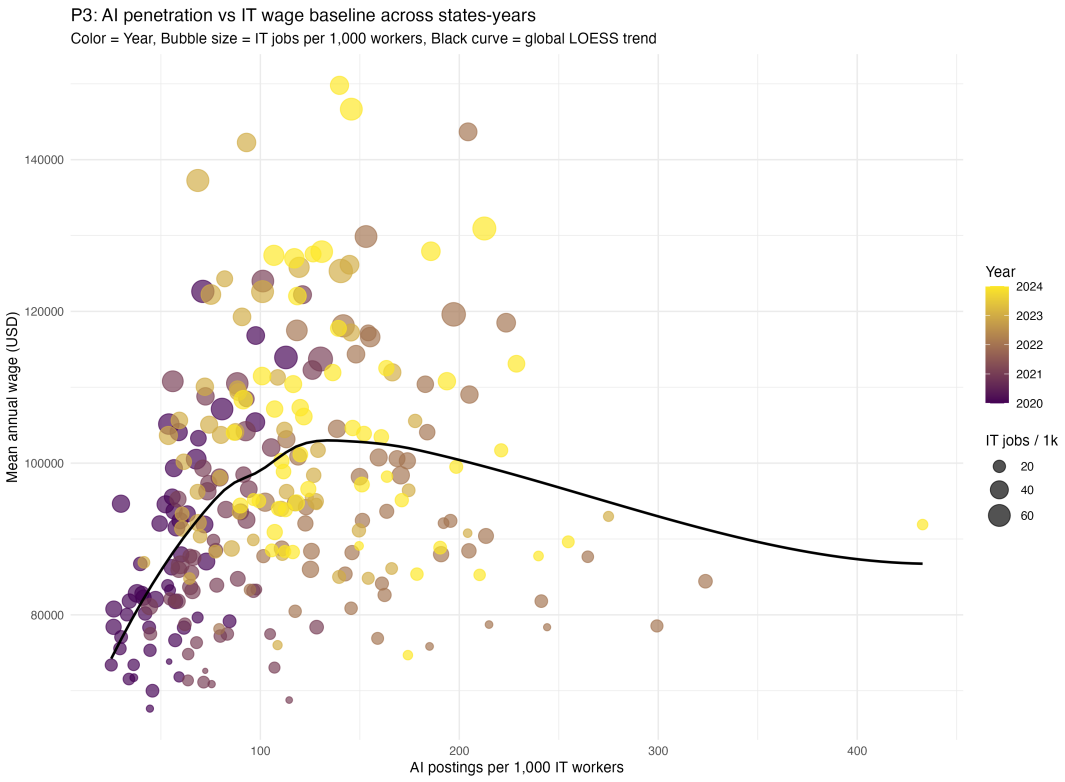


The main pattern is straightforward: most states move upward in nominal wages, and high-penetration states generally remain in higher wage bands across both endpoints.

Presentation P3: Visually Impressive Multi-Variable Display

For P3, I include a multivariable bubble plot with four mapped dimensions:

- x: AI postings per 1,000 IT workers;
- y: mean annual wage;
- bubble size: IT jobs per 1,000 workers;
- color: year (2020–2024), plus LOESS trend.



This figure shows an upward but nonlinear trend, with larger job-density bubbles more concentrated in higher-wage areas. It communicates cross-sectional and time information in the same panel without overloading the reader.

From a presentation perspective, this plot is useful because the reader can read level, size, and time at a glance. That helps connect the premium question to market depth and temporal change in one view, instead of splitting the narrative across multiple smaller charts.

Presentation P4: Reproducibility Plan

For P4, I summarize the reproducibility steps as a practical checklist:

- Source URLs and extraction notes are versioned in the project manifest.
- Entire pipeline is scripted; no manual spreadsheet edits.
- Randomness is fixed by `set.seed(847)`.
- Output artifacts are written with stable names.
- Updating to future years requires appending source-year rows then rerunning the pipeline.

Conclusion

Across exploration, modeling, feature extraction, and presentation views, the evidence supports a restrained conclusion: **AI demand penetration is associated with higher observed wage environments in raw comparisons**, but once state and year heterogeneity are controlled, the standalone within-state premium signal is weak in this panel period. So the strongest claim is not "AI has no premium"; instead, it is that state-level premium effects are conditional, heterogeneous, and partly absorbed by broader labor-market structure.

This is still a meaningful result for policy and strategy. At the same time, a descriptive check in this panel shows that **total Computer and Mathematical employment rises from 2020 to 2024 rather than falling**. For workforce planners, AI-related pay pressure may appear strongest in already deep IT ecosystems.